

Machine Learning Project: Student Performance Prediction

Course/

Applications of Artificial Intelligence (CIT 1403)

Dataset Description

Dataset Overview

- Dataset Name: student_performance_updated_1000.csv
- Instances: 1,000 student records
- Attributes: 11 features including demographic, academic, and behavioral data
- Target Variable: FinalGrade (converted to binary classification: Pass/True vs Fail/False)

Attributes Used

- Nominal Attributes: Gender, ParentalSupport, Online Classes Taken
- Numeric Attributes: AttendanceRate, StudyHoursPerWeek, PreviousGrade, ExtracurricularActivities, Study Hours, Attendance (%)
- Removed Attributes: StudentID, Name (irrelevant for prediction)

Preprocessing Steps

WEKA Preprocessing Pipeline

1. Data Loading: Imported CSV file into WEKA Explorer
2. Attribute Removal: Removed StudentID and Name using Remove filter
3. Missing Values: Handled using ReplaceMissingValues filter
4. Normalization: Applied Standardize filter to normalize numeric data
5. Data Conversion: Converted nominal attributes using automatic WEKA handling

Preprocessing Impact

- Ensured all algorithms received clean, normalized data
- Prevented bias from missing values
- Improved convergence for distance-based algorithms

Algorithm Descriptions

J48 (Decision Tree)

- Core Idea: Builds a tree structure by selecting attributes that maximize information gain at each node
- Selection Reason: Highly interpretable, handles both numeric and nominal data well
- WEKA Parameters: Default settings (-C 0.25 -M 2)

Naive Bayes

- Core Idea: Probabilistic classifier based on Bayes' theorem with feature independence assumption
- Selection Reason: Fast training, works well with small datasets, handles missing values naturally
- WEKA Parameters: Default settings

Random Forest

- Core Idea: Ensemble method that builds multiple decision trees and aggregates their predictions
- Selection Reason: Reduces overfitting, robust to noise, often provides best performance
- WEKA Parameters: 100 iterations, 100 trees, default other settings

Model Evaluation & Comparison

Evaluation Method

- 10-Fold Cross Validation: Ensures reliable performance estimation
- Metrics Reported: Accuracy, Precision, Recall, F-Measure, ROC Area, Confusion Matrix

Performance Table

Metric	J48	Naive Bayes	Random Forest
Accuracy	49.23%	50.36%	51.59%
Precision	0.484	0.501	0.515
Recall	0.492	0.504	0.516
F-Measure	0.456	0.487	0.500
ROC Area	0.488	0.501	0.514
Kappa	-0.0235	0.0017	0.0265

Confusion Matrices

J48:

```
a    b  <-- classified as
```

```
110  370 |  a = False
```

```
125  370 |  b = True
```

Naive Bayes:

```
a    b  <-- classified as
```

```
155  325 |  a = False
```

```
159  336 |  b = True
```

Random Forest:

```
a    b  <-- classified as
```

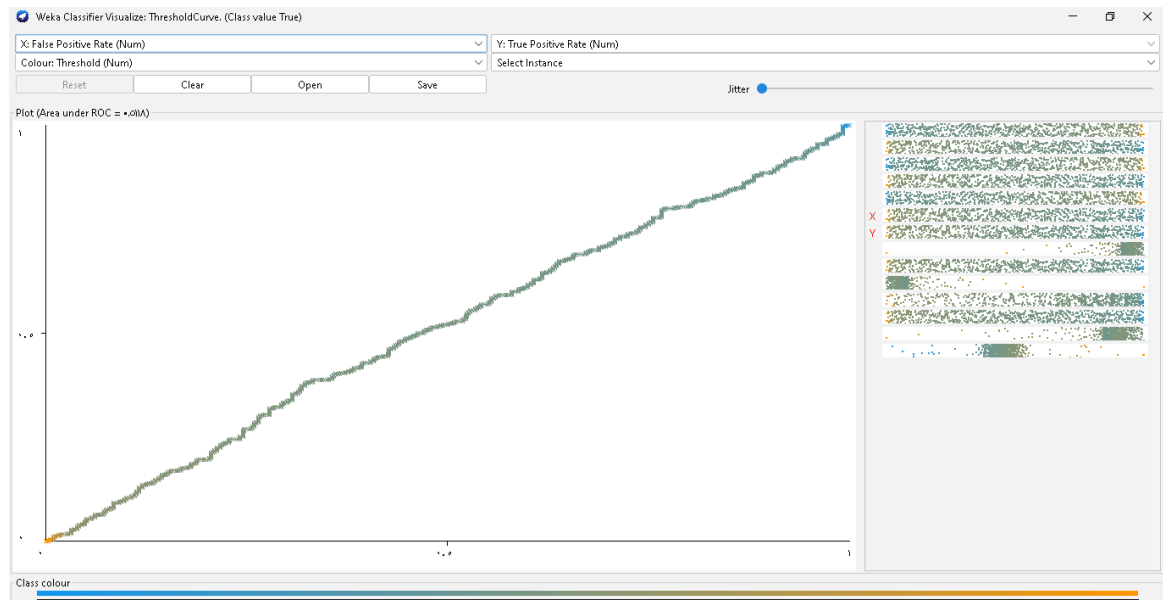
```
162  318 |  a = False
```

```
154  341 |  b = True
```

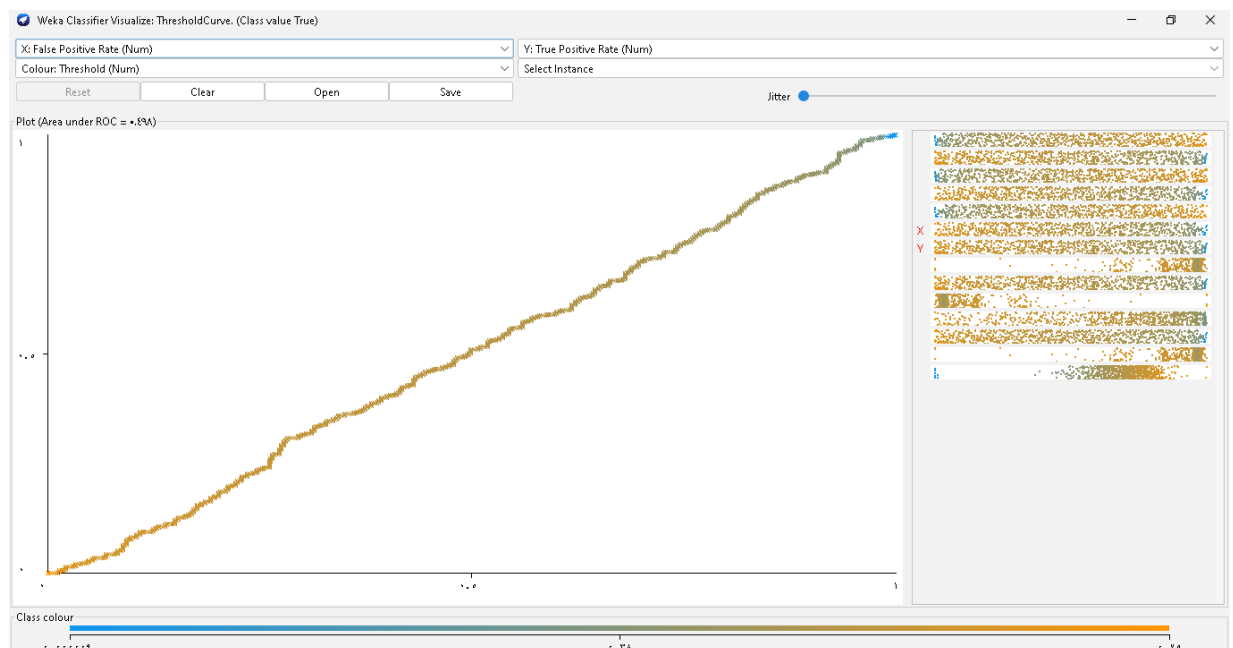
Visualization Analysis

ROC Curve Comparison

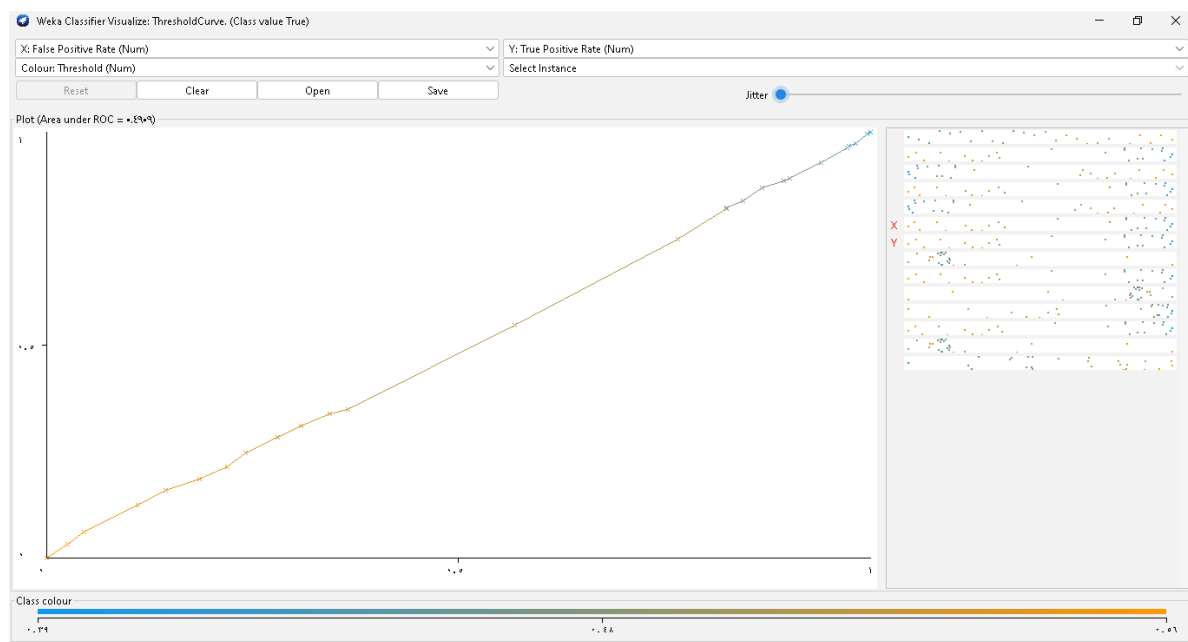
Random Forest:



Naive Bayes:

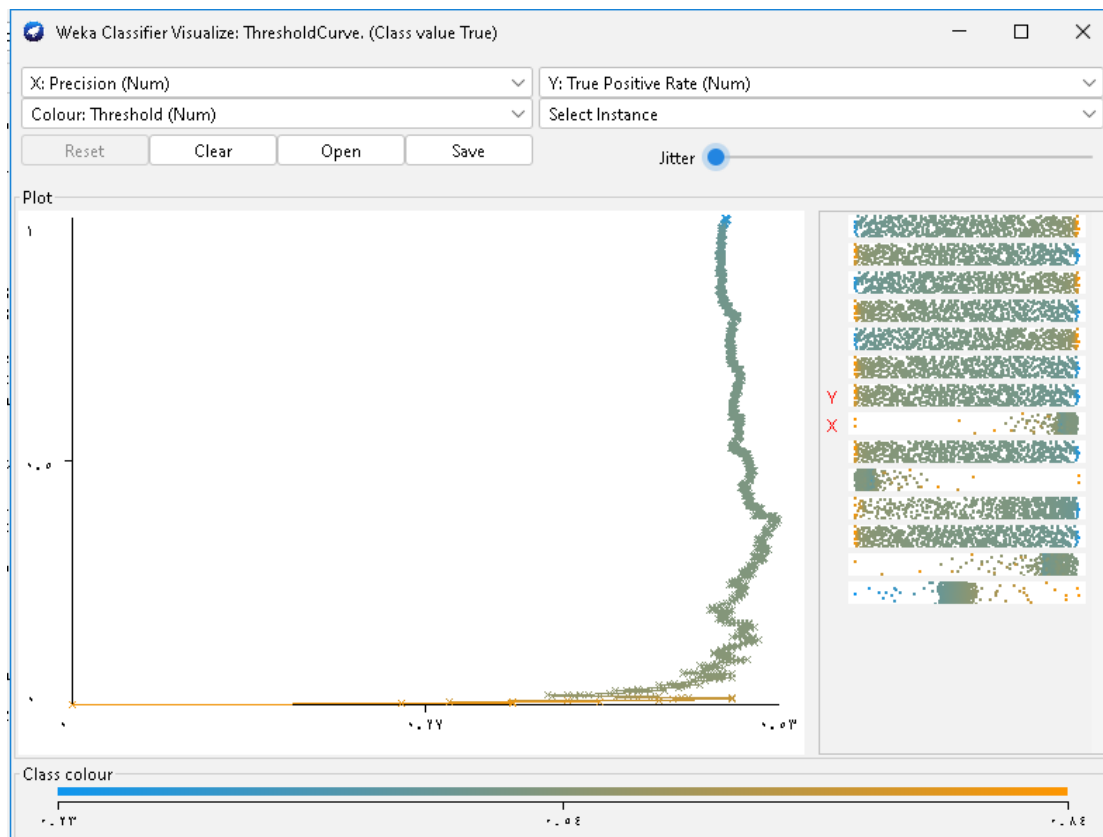
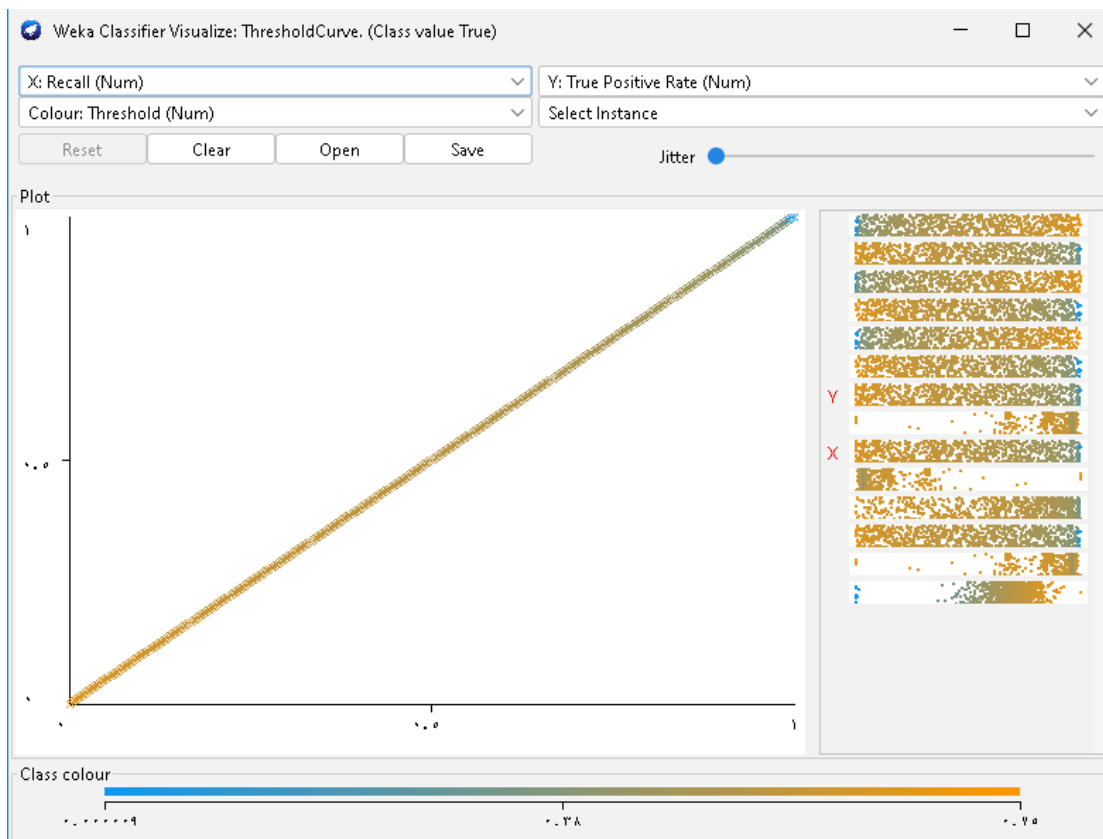


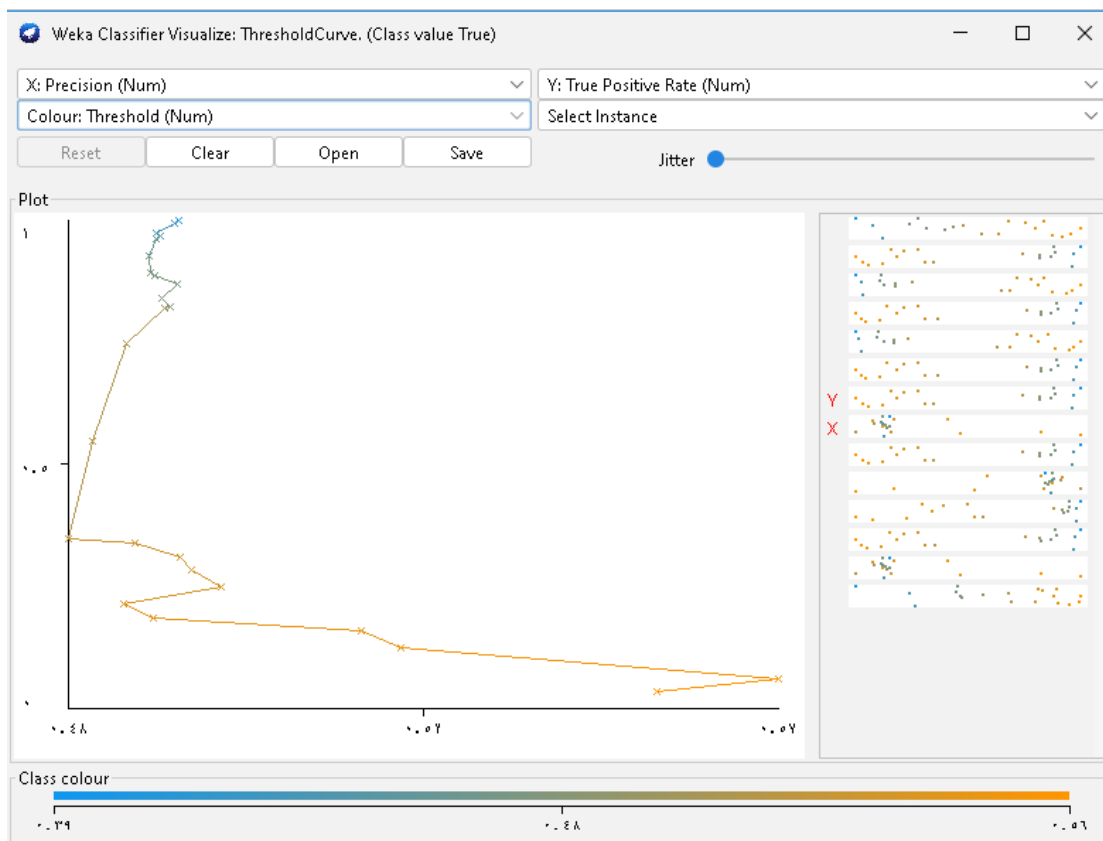
J48:



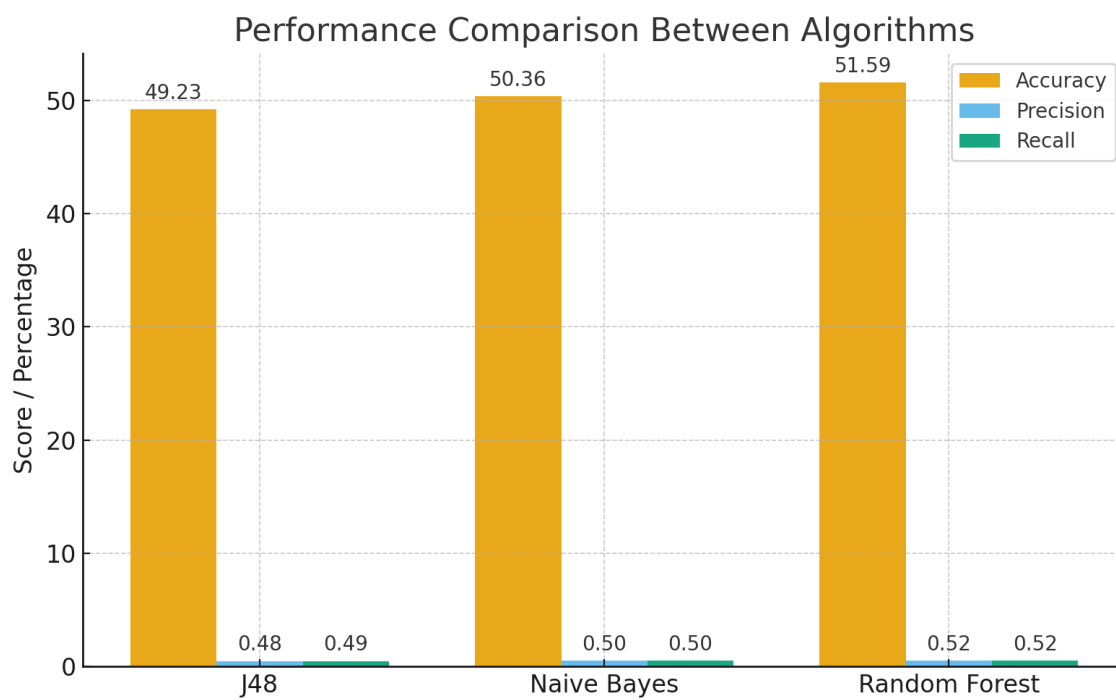
- All algorithms show ROC areas close to 0.5, indicating limited discriminative power
- Random Forest shows slight improvement over others

Performance Metrics Bar Chart





Bar chart:



- Random Forest consistently outperforms other algorithms across all metrics
- All models show modest performance, suggesting feature limitations

Decision Tree Visualization (J48)

ExtracurricularActivities ≤ 0.4685

| ParentalSupport = High

| | StudyHoursPerWeek ≤ 0.7148 : True (236.0/105.0)

| | StudyHoursPerWeek > 0.7148 : False (43.0/18.0)

| ParentalSupport = Medium: False (259.0/125.0)

| ParentalSupport = Low: True (234.0/108.0)

ExtracurricularActivities > 0.4685 : False (203.0/95.0)

Analysis and Discussion

Best Performing Algorithm

Random Forest achieved the best performance (51.59% accuracy) due to:

- Ensemble learning reducing overfitting
- Handling of feature interactions better than single decision tree
- Robustness to noise in the dataset

Preprocessing Impact

- Standardization improved performance for all algorithms
- Missing value handling prevented data loss
- Feature selection (removing irrelevant attributes) improved model focus

Pattern Insights

- J48 Tree Structure reveals that ExtracurricularActivities and ParentalSupport are key decision points
- Confusion Matrices show all models struggle with False class identification
- Low ROC Areas suggest weak predictive power with current features

Challenges and Limitations

1. Data Quality: Missing values and potential inconsistencies in attributes like "Study Hours" vs "StudyHoursPerWeek"
2. Feature Relevance: Current attributes may not strongly predict academic performance
3. Class Imbalance: Models show bias toward predicting "True" class
4. Model Performance: All algorithms show near-random performance, indicating need for feature engineering

Conclusion

Key Findings

- Random Forest performed best but with modest accuracy (51.59%)
- Current features have limited predictive power for student performance
- Preprocessing improved model stability but couldn't overcome fundamental feature limitations

Recommendations for Improvement

1. Feature Engineering: Create composite features (e.g., study efficiency, attendance consistency)
2. Data Collection: Gather more relevant features (e.g., socioeconomic status, learning style, course difficulty)
3. Algorithm Tuning: Experiment with hyperparameter optimization
4. Alternative Approaches: Try clustering or regression instead of classification